

Statistics Practice Set — S31

Probability, Conditional Probability, Bayes & Business Analysis

No hints. Focus on calculation + interpretation.

Q1 — Conditional Probability: Customer Purchase

Source	Customers	Purchased
Advertisement	1,200	240
Organic	1,800	360

1. Find total customers.
 2. Find total purchasers.
 3. Find $P(\text{Purchased} \mid \text{Advertisement})$.
 4. Find $P(\text{Advertisement} \mid \text{Purchased})$.
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Q2 — Independent vs Dependent

A company records:

	Purchased	Did Not Purchase
Saw Ad	200	800
Did Not See Ad	100	900

1. Find purchase rate among customers who saw the ad.
 2. Find purchase rate among customers who did not see the ad.
 3. Are the two events independent or dependent?
 4. Explain why.
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Q3 — Addition Rule: Customer Channels

A company has **2,000 customers**:

- 1,100 use the App
- 900 use the Website
- 400 use both

Find:

1. Probability of using the App.
 2. Probability of using the Website.
 3. Probability of using at least one channel.
 4. Number of customers using neither.
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Q4 — Complement: Churn

A subscription company has a churn probability of **7%**.

1. Find the probability of **not churning**.
 2. Out of 5,000 customers, how many are expected not to churn?
 3. Is this exact number guaranteed? Explain.
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Q5 — Mutually Exclusive vs Overlapping

A die is rolled.

- A = getting a number **less than 4**
- B = getting an **even number**
- C = getting a number **greater than 3**

Find:

1. $A \cap B$
2. Are A and B mutually exclusive?
3. $B \cap C$
4. Are B and C mutually exclusive?

Q6 — Bayes: Fraud Detection

Transaction Type	Transactions	Fraud
Online	8,000	240
Offline	4,000	60

Find:

1. Overall fraud probability.
2. $P(\text{Fraud} \mid \text{Online})$
3. $P(\text{Online} \mid \text{Fraud})$
4. Explain why the denominators in Q2 and Q3 are different.

Q7 — Rate vs Count: Marketing

Campaign	Reached	Purchased
A	6,000	480
B	2,000	280

1. Find the conversion rate for A.
2. Find the conversion rate for B.
3. Which campaign has more purchasers?
4. Which campaign has the better conversion rate?

Q8 — Find the Wrong Analyst Interpretation

An analyst says:

“Campaign A generated more purchases than Campaign B, so Campaign A definitely performed better.”

Do you agree or disagree?

Give **one reason** using the data from Q7.

Q9 — Reverse Probability: Employee Skills

Department	Employees	Python Skilled
Data Analytics	300	180
Finance	500	100
Marketing	700	70

Find:

1. Total employees.
2. Total Python-skilled employees.
3. $P(\text{Python Skilled} \mid \text{Data Analytics})$
4. $P(\text{Data Analytics} \mid \text{Python Skilled})$

Q10 — Full Data Analyst Case: Churn

Plan	Customers	Churned
Basic	4,000	520
Premium	2,000	180
Enterprise	1,000	50

1. Find the churn rate for each plan.
2. Which plan has the highest churn rate?
3. Can you say the Basic plan **causes** churn? Explain briefly.